

GH

Human Growth Hormone (Somatropin / rHGH)

GROWTH HORMONE

BODY RECOMP

FAT LOSS

ANTI-AGING

RECOVERY

WHAT IS IT?

Human Growth Hormone (HGH/somatropin) is a 191-amino-acid peptide hormone secreted by the anterior pituitary. Pharmaceutical GH (recombinant human growth hormone / rHGH) is identical in structure to endogenous GH. It is used clinically for GH deficiency, short stature, HIV wasting, and Turner syndrome. In performance and anti-aging medicine, it is used for body recomposition, fat loss, recovery, and longevity.

DOSING AT A GLANCE

Anti-aging / TRT	Body recomp	Performance	Advanced split
1–2 IU/day	2–4 IU/day	4–6 IU/day	2–3x/day

KEY BENEFITS

- Powerful lipolysis — promotes fat mobilization, especially visceral and abdominal fat
- Increases IGF-1 — drives muscle protein synthesis and recovery
- Significantly improves body composition (fat loss + lean mass over 3–6+ months)
- Enhanced recovery — reduces muscle soreness, accelerates tissue repair
- Improved sleep quality (endogenous GH is primarily released during deep sleep)
- Anti-aging: improved skin elasticity, collagen, hair quality
- Increased bone density with long-term use
- Improved insulin sensitivity at low doses (reversed at high doses)
- Enhanced cognitive function and energy
- Joint and connective tissue repair and maintenance

HOW IT WORKS

GH acts on GH receptors in the liver to stimulate IGF-1 production, which drives anabolic effects in muscle and bone. Directly, GH activates lipolysis via hormone-sensitive lipase in adipocytes and counteracts insulin's actions in fat tissue. GH also upregulates protein synthesis independently of IGF-1 via JAK2-STAT5 signaling in muscle.

PROTOCOLS

- **Anti-Aging / Wellness:** 1–2 IU SC daily before bed. Monitor IGF-1 every 3 months.
- **Body Recomp:** 2–3 IU SC daily (morning fasted or split AM/PM).
- **Performance:** 4–6 IU/day split into 2 injections (pre-training + before bed).
- **Cycle length:** Minimum 3 months for meaningful body composition changes; 6+ months optimal.
- **Stack:** GH + testosterone + insulin (advanced) = elite muscle-building protocol.
- **Peptide alternative:** CJC-1295 + Ipamorelin stimulates endogenous GH at lower cost.

SIDE EFFECTS

- Water retention and edema (dose dependent — reduces over weeks)
- Carpal tunnel syndrome (tingling/numbness in hands — reduce dose if severe)
- Joint pain (arthralgias) — common early, usually transient
- Insulin resistance at higher doses — monitor fasting glucose
- Headache (intracranial pressure — rare at therapeutic doses)
- Potential organ growth (gut distension) at high sustained doses
- Theoretical cancer promotion risk (IGF-1 is mitogenic) — avoid in active cancer

STORAGE

Lyophilized vials: refrigerate 2–8°C. Reconstituted: refrigerate, use within 21–28 days. Never freeze reconstituted GH.

■ **Pharmaceutical GH is expensive (\$300–800+/month). Verify product via HPLC — counterfeit and underdosed GH is extremely common. Monitor IGF-1, fasting glucose, HbA1c every 3 months.**

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